

Michele Stingo

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Summary

As a Ph.D. in Computational Linguistics, my expertise lies at the intersection of Natural Language Processing, Large Language Models, and the design of intelligent agent-based systems. My current research focuses on the development of cognitive agents and multi-agent architectures, with particular attention to orchestration mechanisms, stateful reasoning, and coordination patterns implemented through frameworks such as LangGraph.

A central component of my research concerns knowledge representation and reasoning, particularly the use of ontologies and graph-based models to structure and organize information within intelligent systems. I have practical experience with tools such as Protégé and Neo4j, and I explore how formal knowledge structures can enhance the capabilities of LLM-based architectures. These concepts increasingly inform my work on agent memory, shared state management, and symbolic-subsymbolic integration, enabling the design of systems in which language models interact with structured knowledge to support more reliable reasoning and decision-making.

I specialize in the use of advanced prompt engineering techniques to fully leverage the reasoning, planning, and interaction capabilities of modern language models. I design prompts and control flows that support complex decision-making processes, tool usage, and inter-agent communication. My work actively contributes to transforming LLMs from simple text generators into goal-oriented components within structured systems.

I work comfortably across several development environments, including Jupyter, Colab, Kaggle Notebooks, and Anaconda-based ecosystems. My previous experience includes the development of conversational systems using platforms such as IBM WatsonX and Azure Cognitive Services, providing a strong foundation in production-oriented AI solutions. I actively explore and evaluate emerging frameworks and tools for LLM-based applications, including LangChain, LangGraph, Ollama, and the Hugging Face Transformers ecosystem. My work is guided by continuous research into new scientific directions, architectural patterns, and methodologies aimed at building robust, interpretable, and scalable intelligent systems.

Experience

Activa Digital, Senior Research and Development Scientist

Rome, Italy

Sept 2023 – present
2 years 9 months

- Developing cognitive agents and multi-agent architectures with a focus on orchestration and coordination patterns using LangGraph.
- Designing systems where language models interact with structured knowledge (ontologies and graph-based models) to enhance reasoning.
- Utilizing tools like Protégé and Neo4j for knowledge representation and symbolic-subsymbolic integration.
- Implementing advanced prompt engineering and control flows to support complex decision-making and tool usage in LLM-based systems.
- Leading R&D team efforts and collaborating on grant procurement and experimental design.

Luo, Computational Linguist, Software Analyst, and Front-End Developer

Rome, Italy

Aug 2017 – Sept 2023
6 years 2 months

- Performed linguistic data annotation and created domain-specific monolingual corpora.
- Evaluated NLP tools including lemmatizers, part-of-speech taggers, and syntactic parsers.
- Developed and manipulated ontological resources using OWL, XML, and JSON.
- Developed mobile native applications using Flutter and web applications using Angular.

University of Salerno, PhD Researcher and Teaching Assistant

Salerno, Italy

Oct 2016 – Mar 2020
3 years 6 months

- Conducted teaching activities and mentored graduate and undergraduate students.
- Enhanced AI industrial pipelines and improved pragmatic aspects of NLP solutions.
- Created linguistic corpora and electronic dictionaries.

Education

PhD	University of Salerno , Computational Linguistics Thesis: La formalizzazione della conoscenza linguistica in ambito industriale (Techniques, algorithms and electronic resources for Big Data for Multi-Agent Specialized Systems).	Salerno, Italy Oct 2016 – Mar 2020
MA	Ca' Foscari University of Venice , Language Sciences - Computational Linguistics - Final Grade: 110/110 cum laude. - Thesis: Computational Analysis on a Corpus of Political Satire Articles.	Venice, Italy Sept 2013 – Mar 2016
BA	University of Salerno , Literature and Philosophy - Modern Literature Final Grade: 110/110 cum laude.	Salerno, Italy Jan 2009 – Jan 2013
Diploma	T.C. Guido Dorso , High School Diploma	Sarno, Italy Jan 2006

Publications

xch-MIND: Multi-Agent Interpretive Knowledge Graph Augmentation over Cultural Heritage Linked Data <i>Michele Stingo</i> (Proceedings of the 5th Workshop on LLM-Integrated Knowledge Graph Generation from Text (Text2KG) at the 23rd European Semantic Web Conference (ESWC))	Jan 2026
Mina: SeMantic vIrtual Assistant for Domain oNtology Based Question-Answering N. Fiore, G. Parente, <i>M. Stingo</i> , M. Polito (Springer International Publishing (IncOs-2020))	Jan 2020
Assisted Contracts Validation through LCS problem resolution F. Martinucci, <i>M. Stingo</i> , M. Polito (AIDAinformazioni)	Jan 2020
Automatic text preprocessing for intelligent dialog agents A. Maisto, S. Pelosi, M. Polito, <i>M. Stingo</i> (Springer International Publishing (Waina-33))	Jan 2019
Multi-Word Expressions in spoken language: PoliSdict D. Trotta, <i>M. Stingo</i> , R. Guarasci, A. Teresa, A. Elia (CLiC-it 2018)	Jan 2018
OPI: Ontologia dei prodotti italiani con denominazione DOP, IGP, STG <i>M. Stingo</i> (AIDAinformazioni)	Jan 2018
A hybrid method for the extraction and classification of product features from user-generated contents A. Maisto, S. Pelosi, <i>M. Stingo</i> , R. Guarasci (Lingue e Linguaggi)	Jan 2017
Detecting satire in italian political commentaries <i>M. Stingo</i> , R. Delmonte (NLPMJ - AAAI)	Jan 2016

Skills

Languages: Italian (Native), English (C1), French (B2), Spanish (B1), Albanian (A2)

AI & NLP: LLMs, LangChain, LangGraph, Ollama, Hugging Face Transformers, NLTK, Sentiment Analysis, POS Tagging

Knowledge Engineering: Ontologies (OWL), Graph Databases, Neo4j, Protégé, Description Logic, XML, JSON

Software Development: Python, Prolog, Flutter, Angular, HTML5, CSS3, Bootstrap, Jupyter, Anaconda

Digital Competencies: Information and data literacy, communication and collaboration, digital content creation, problem solving

New Section
